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Complexity and the Transition to Post-Quantum Security: Cryptographic Challenges regarding Shor's and Grover's Algorithms

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Abstract

Quantum computing has become one of the most important emerging technologies, with the potential to compromise traditional cryptographic systems such as RSA and ECC, which are widely used for information security. This study proposes a strategic roadmap to help organizations transition to post-quantum security systems, mitigating the risks generated by advanced quantum algorithms like Shor's and Grover's. The methodology includes a review of scientific literature and a comparative analysis of emerging cryptographic technologies, including lattice-based cryptography, code-based cryptography, and quantum key distribution (QKD). The results allowed evidence of the need for proactive and structured planning, including a gradual implementation of post-quantum cryptographic solutions, updating security policies, and specialized training programs for technical and executive personnel. This paper provides a comprehensive framework to help organizations anticipate quantum threats and ensure the resilience of their cryptographic security infrastructure.

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1. Introduction

Quantum computing has become one of the most disruptive technological innovations, potentially revolutionizing several fields, especially cryptography, and cybersecurity. Quantum Computing also represents a growing threat to traditional cryptography techniques due to its ability to break conventional security systems [1].

The security of systems using techniques like Rivest-Shamir-Adleman (RSA) and Elliptic-curve cryptography (ECC) is being undermined due to the ability of Quantum algorithms to factor large numbers efficiently. The arrival of quantum computing represents a new challenge for organizations in ensuring the security of their systems and the confidentiality of their data [2]. Understanding the impact of quantum computing on information security and cryptography is essential, and it is necessary to identify the differences between traditional computing and quantum computing [3]. This understanding will enable a comprehensive evaluation of emerging threats and the opportunities presented by these new technologies, as well as the development of effective strategies to adopt cryptographic solutions capable of withstanding the power of quantum computers [3].

In traditional computing, bits are used to process information. A bit can take values of either 0 or 1. These bits are operated by logic gates within computer processors, enabling the execution of instructions to solve specific problems [4]. For example, a processor using a Quicksort algorithm to sort a list of numbers must follow a sequential set of steps, comparing and rearranging pairs of numbers in each iteration. Computational complexity is measured by the number of operations required to complete the task, which is averaged $O\ (n\ log\ n)$ in this case.

On the other hand, quantum computing employs qubits instead of bits. Due to the superposition principle, a qubit can simultaneously represent values of 0 and 1. Qubits can be entangled, which means that the state of one qubit can be instantaneously correlated with another, regardless of the distance between them. This property enables quantum computers to perform large-scale parallel operations and, depending on the problem, potentially perform far more powerfully and efficiently than traditional computers [4].

One of the most significant risks arises from Shor's algorithm (1994)[5], which allows for efficient factorization of integers and the calculation of discrete logarithms exponentially faster than any known classical algorithm. This breakthrough threatens the security of most contemporary cryptographic protocols, including the Public Key Infrastructure (PKI) that underpins financial transactions, government communications, and other critical systems.

Similarly, Grover's algorithm (1996) [6] accelerates searches in unstructured databases, significantly weakening symmetric encryption schemes such as the Advanced Encryption Standard (AES). Traditional cryptography relies on the computational infeasibility of performing many sequential calculations to decrypt a key. For instance, attempting all possible combinations through brute-force encryption is impractical with classical computing due to the time required. However, qubits allow a quantum computer to evaluate multiple combinations simultaneously, rendering specific traditional encryption methods vulnerable [7]. Quantum superposition will enable qubits to exist in various states simultaneously, meaning that a quantum computer can process an exponential number of possible states in parallel, directly affecting how computational problems are approached [8].

Post-quantum cryptography is focused on developing algorithms based on mathematical problems that remain challenging even for quantum computers. This paper aims to establish a roadmap that encompasses theoretical foundations for the practical application of post-quantum security solutions, ensuring the continuity and security of operations in the short, mid, and long term and preparing organizations for the transition to post-quantum security systems, ensuring confidentiality, integrity, and authenticity of data. This paper will explore technical and operational challenges in implementing quantum and post-quantum cryptography technologies and propose a roadmap to guide this transition.

This article is organized as follows: Section 2 outlines the analytical and experimental methods used to analyze the current state of quantum cryptography and the development of the proposed framework. Section 3 shows the literature review results, focusing on Shor's and Grover's algorithms, two of the most important post-quantum algorithms. Finally, section 4 presents the main conclusions and the challenges associated with transitioning to a quantum-resistant security environment.

2. Methodology

This paper adopts a research methodology that follows a structured approach divided into two main stages. The first stage included a literature review and bibliometric analysis of Shor's and Grover's algorithms. The search was carried out in the Scopus database with the query:

TITLE-ABS-KEY (”quantum computing” OR ”quantum information”) AND (”post-quantum algorithm*” OR ”post-quantum cryptograph*” OR ”Shor* algorithm” OR ”Grover* algorithm”).

The resulting documents were limited to peer-reviewed journals and conference papers written in English. A co-occurrence analysis using VosViewer with the most frequent terms and keywords was performed. The last stage included a descriptive process for developing a roadmap for evaluating cryptographic infrastructure and guiding organizations through the transition process to post-quantum cryptography. This roadmap is divided in several stages aiming to assist organizations in moving from the early stages of awareness to the more advanced stages, including monitoring and continuous improvement.

3. Results

3.1. Literature Review

3.1.1. Shor's Algorithm: Significance and Advances

Shor's algorithm has become a milestone in quantum computing, due to its ability to efficiently factorize integers in polynomial time, posing a direct hazard to traditional cryptographic systems such as RSA and ECC [9, 10, 11, 12]. Recent studies have focused on implementing Shor's algorithm for improving the efficiency of quantum circuits and adapting it to distributed quantum computing environments, thereby enhancing its scalability and applicability [9, 13, 14].

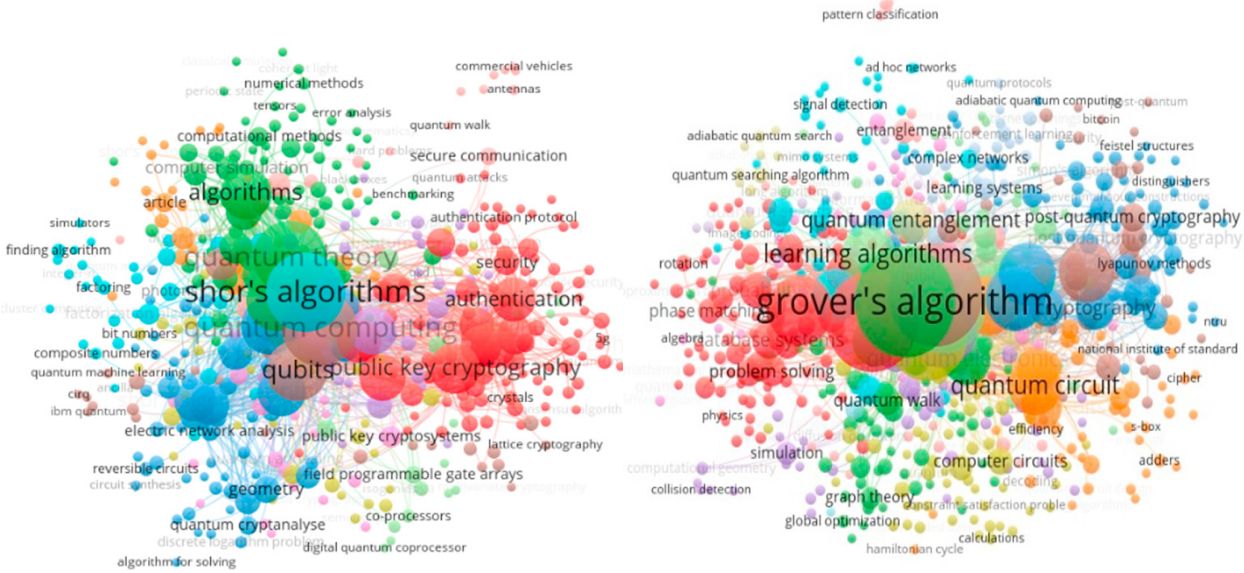
Furthermore, scientific literature has explored the optimization of the Quantum Fourier Transform (QFT), as a critical component of the algorithm, to increase the accuracy of quantum calculations and reduce the number of required gates [15, 16]. This advance redefines quantum factorization and increases the need for cryptographic systems to resist quantum attacks. [12]. Several operational and technical issues limit the practical application of Shor's algorithm despite the notable advancements in its implementation. One of the primary issues is the influence of quantum noise, which can severely affect computation precision and restrain the algorithm's capacity to factor big numbers [11].

On the other hand, optimizing quantum circuits is another crucial challenge. Recent studies have focused on minimizing cumulative errors and improving computational efficiency, reducing the number of gates (particularly CNOT gates) [9, 13]. Moreover, the algorithm's scalability in distributed environments raises questions regarding the synchronization and resource management in multi-node quantum computers [14]. Large-scale simulations have revealed difficulties in the practical implementation of the algorithm, even when using advanced architectures such as GPUs to emulate quantum behaviors [15]. An emerging research area involves assessing the resilience of traditional cryptographic security systems against quantum algorithms. These advancements are essential for developing transition strategies towards post-quantum cryptography [12].

The importance of Shor's algorithm in integer factorization and discrete logarithm problem-solving is highlighted by a co-occurrence analysis of the most used keywords in scientific literature using VOSviewer. Relationships to concepts like quantum theory, qubits, and quantum computing suggest a strong emphasis on real-world application and optimizing quantum circuits for effective operation. The mapping also highlights the danger it poses to existing cryptographic systems by exposing its use in crucial domains like security, authentication, and secure communication. Decoherence, error analysis, and reversible circuit synthesis are among the technical obstacles that must be overcome to implement the algorithm in a scalable and accurate manner in the upcoming era of quantum computing (see Figure 1a).

3.1.2. Grover's Algorithm: A Quantum Procedure for Unstructured Search

Grover's algorithm is a quantum procedure that provides a quadratic speedup for unstructured search in databases, making it a crucial component within the quantum computing paradigm [17, 18, 19, 20, 21]. Unlike traditional algorithms, which require a linear number of evaluations $O(N)$ to find an element in an unordered list, Grover's algorithm



(a) Shor's Algorithm (b) Grover's Algorithm

Fig. 1: Bibliometric analysis of Shor's and Grover's algorithms using VOSviewer

achieves this goal in $O(\sqrt{N})$ operations, offering an exponential improvement in efficiency [22, 23]. This algorithm relies on amplitude amplification, a quantum technique that increases the probability of observing the desired state, allowing search problems to be solved more efficiently than with any classical algorithm [24, 25, 26]. Beyond database search, Grover's algorithm has proven useful in combinatorial optimization, cryptographic collision verification, and constraint satisfaction problems [17, 18]. These characteristics position Grover's algorithm as both a challenge and an opportunity to redefine computational efficiency in the quantum era [19, 20].

Despite its potential, the practical implementation of Grover's algorithm faces significant technical challenges, particularly related to quantum noise and decoherence errors that degrade computational accuracy [17, 18, 19, 20]. These problems arise mainly in current quantum hardware architectures, where qubits are susceptible to environmental interference, affecting the algorithm's stability and reliability [21, 22]. Furthermore, optimizing the quantum circuits that implement Grover's algorithm is crucial for reducing computational complexity and the number of logical gates involved, as an excess of operations increases the rate of cumulative errors [25, 26]. Another major challenge is the algorithm's scalability—translating its theoretical efficiency to distributed quantum systems requires coordinating multiple qubits in parallel, adding an additional layer of complexity [23, 24]. There are also challenges in applying Grover's algorithm to post-quantum cryptography, where its feasibility for cracking symmetric cryptography systems and guaranteeing security against quantum attacks needs to be confirmed [25, 26].

These difficulties show that to maximize Grover's algorithm's implementation and ensure its efficacy in actual quantum environments, it is necessary to enhance error correction methods and create fresh approaches [17, 18, 19]. A co-occurrence analysis of keywords related to Grover's algorithm using VOSviewer highlights its importance in unstructured search with quadratic speedup and its expanding use in crucial fields like post-quantum cryptography, global optimization, and signal detection. Relationships with concepts like learning algorithms, quantum circuits, and quantum entanglement denote an emphasis on real-world application and solving complex problems (see Figure 1b).

3.2. Roadmap

Developing a resilient roadmap for post-quantum security must consider a wide process from the early stages of the lack of awareness to the late stages of evaluation, risk-based prioritization, cautious selection and integration of quantum-resistant algorithms, and continuous adaptation. Organizations must combine organizational preparedness with technological advancements to guarantee a safe and robust transition to the post-quantum era. The proposed

Table 1: Proposed Roadmap to ensure an effective transition to post-quantum cryptography.

Stage	Recommendations
Awareness and Training	· Identify and train multidisciplinary teams on the fundamentals of post-quantum cryptography. · Disseminate information about the risks associated with quantum computing.
Cryptographic Asset Inventory	· Catalog all systems using vulnerable algorithms such as RSA, ECC, or DSA. · Prioritize assets based on their criticality and impact in case of failure.
Evaluation of Post-Quantum Algorithms	· Analyze the alternatives recommended by NIST, such as Kyber and Dilithium, evaluating their performance in real-world scenarios. · Consider implementing hybrid solutions that combine classical and post-quantum cryptography.
Pilot Testing	· Conduct controlled tests in simulated environments to evaluate the compatibility of selected solutions with existing systems. · Adjust configurations based on the obtained results.
Gradual Migration	· Introduce post-quantum solutions in less critical systems to minimize operational risks. · Scale implementation to priority systems as the technologies are validated.
Continuous Monitoring	· Establish a monitoring system to evaluate the performance of new algorithms. · Update solutions as international standards and technological threats evolve.

Table 2: Assessment of Current Cryptographic Infrastructure

Process Step	Description
Identification of Critical Algorithms	Catalog current cryptographic algorithms in use, such as RSA (1024, 2048 bits), ECC, and AES. Determine the specific vulnerability of each algorithm to quantum attacks, such as Shor's and Grover's algorithms.
Risk Classification	Evaluate the exposure of digital assets to potential quantum attacks. Classify risks into low, medium, and critical levels based on data sensitivity and the expected impact of a failure.
Performance Testing	Conduct simulations to assess processing time and resources required to implement post-quantum solutions. Compare the effectiveness of emerging algorithms like Kyber and Saber against current standards.
Comparative Performance Analysis	Measure the impact of post-quantum algorithms on latency and resource consumption in current systems. Identify technical limitations that may arise during the transition, such as hardware incompatibility.
Review of Standards and Regulations	Compare the current cryptographic infrastructure with the standards recommended by NIST. Analyze international best practices to guide specific recommendations.

roadmap develops a framework divided into six key stages to ensure an effective transition to post-quantum cryptography, as shown in Table 1.

3.3. Assessment of Current Cryptographic Infrastructure and Identification of Cryptographic Assets

Table 2 shows the proposed process for assessment of the cryptographic infrastructure. The first stage in the transition to post-quantum security systems is to identify all cryptographic assets within the organization. Conducting an exhaustive inventory of all systems and applications using cryptography is essential. It is also necessary to identify the specific versions of the implemented algorithms, considering key lengths (as shorter keys are more vulnerable to quantum attacks). The following elements should be considered at this stage:

- Authentication systems: Identify the use of algorithms such as RSA or ECC for user and device authentication.
- Secure communication protocols: Identify how the data is being transmitted. Identify the use of protocols like SSL/TLS (Secure Sockets Layer/Transport Layer Security)that implement asymmetric cryptography for authentication and key exchange.
- Information Storage Systems: Identify how data storage is secured and how confidential data is protected. Encryption protocols like AES (Advanced Encryption Standard) or other symmetric ciphers should be implemented.
- Digital Signatures and Certificates: These tools help to guarantee the integrity and authenticity of electronic documents and transactions.

3.4. Practical Considerations and Comparisons and Migration Planning

The selection of appropriate post-quantum technology depends on several factors, including performance, compatibility with existing systems, and ease of implementation. Table 3 shows a technical comparison of the

Table 3: Practical Considerations of Post-Quantum and Classical Algorithms

Algorithm	Type	Security	Key Size	Performance	Recommended Use
Kyber	Lattice-based cryptography	High	Medium	High	Data Encryption, Key Exchange
NTRU	Lattice-based cryptography	High	Medium	High	Digital Signatures, Data Encryption
McEliece	Code-based cryptography	Very High	Large	Medium	Applications requiring high security
AES-256	Symmetric Encryption	High	Medium	Very High	Data encryption, Secure Communication
SHA-3	Hash Function	High		High	Digital Signatures, Data Integrity

mentioned algorithms. The transition to post-quantum technologies requires a hybrid implementation strategy. This strategy should combine traditional and post-quantum cryptography to ensure compatibility and security.

- Compatibility: Post-quantum solutions can be implemented alongside traditional technologies. The concept of ”hybridization” allows existing systems to continue functioning while new cryptographic solutions are introduced.
- Implementation Stages: The transition process should be carried out in gradual stages. The process should start with pilot tests in controlled environments. This approach allows for detecting and fixing problems before performing real large-scale operations. Critical systems (financial data, sensitive or personal information related) should be prioritized.
- Real-Time Monitoring: Once new technologies are deployed, it is vital to implement a real-time monitoring system to detect and respond quickly to any security or performance issues.

4. Conclusions

In this paper, a roadmap for the transition to post-quantum security systems has been developed. Scientific literature has shown that quantum algorithms, including Shor and Grover's, may threaten the security of the most used cryptographic methods, such as RSA and ECC. One of the researchers' major conclusions is that even though theage of quantum computing has not matured enough for these attacks to be realized, organizations should not hesitate to prepare. Early planning and awareness are essential to implement a successful roadmap to transition to the post-quantum era.

The threat to data security and privacy is the most apparent and immediate risk, and it has the potential to significantly affect civil society, economic stability, and national security. Significant sociopolitical risks are introduced by quantum computing, particularly in cybersecurity, international power relations, and legal governance. To control these risks and guarantee that quantum technologies benefit society overall, proactive international cooperation, modernized legal frameworks, and ethical standards are crucial.

Although the proposed roadmap is a valuable instrument for directing shifts to a post-quantum era, its limitations may include, depending on the organizational context, a lack of universality, resource limitations, and adaptability. Flexible, context-specific planning and constant adaptation are necessary for a successful transition. There are numerous unresolved technical, organizational, and policy issues with the lengthy and intricate transition process to post-quantum security systems. Various future research opportunities related to algorithm development and cybersecurity have appeared. Developing sector-specific solutions is critical; security analysis and policy coordination will become essential. Addressing these challenges is crucial to ensuring future-proof cybersecurity in the quantum era.

During the preparation of this work, the author(s) used ChatGPT 4o to translate some parts of the document, and Grammarly was used to improve coherence and readability after the translation. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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